

Redrob Challenge Submission

Neural Explainable Candidate Understanding System

TRACK

METHODOLOGY

PERFORMANCE

THE PROBLEM STATEMENT

Traditional recruiter filters overlook high-potential candidates by counting keyword matches rather than modeling role compatibility. The challenge requires building a candidate ranker that behaves like an expert hiring manager.

✕ Keyword Stuffing

Profiles listing dozens of tech skills without actual career timeline evidence easily game search engines.

✕ Synthetic Traps

Over 80 honeypot profiles containing temporal inconsistencies or skill-duration paradoxes pollute the data.

✕ Compute Limits

Ranking 100K candidates must execute locally in under 5 minutes on CPU without using web APIs (OpenAI, etc.).

SYSTEM DESIGN

NEXUS replaces simple keyword searches with a multi-layered cascading evaluation engine. Each layer scoring is computed locally on CPU with zero network dependencies.

Layer 0-2: Fit & Integrity

- Layer 0: Integrity Shield filters honeypots
- Layer 1: JD Parser extracts constraints
- Layer 2: Career DNA tracks trajectory & growth

Layer 3-4: Scoring

- Layer 3: Semantic Scorer uses custom BM25
- Layer 4: Behavioral Scorer groups 23 signals

Layer 5: Decision

- Layer 5: Ensemble Ranker calibrates scores and generates natural explanations

Processing 100K profiles directly through multi-dimensional scoring would violate the 5-minute limit. NEXUS uses a cascading pipeline to optimize execution speed.

Tier 1: Hard Filter

- Checks minimum AI relevance & JD rules
- Funnels 100,000 → 39,012 candidates
- Execution time: 46.7s

Tier 2: BM25 Pre-rank

- High-speed custom BM25 index matching
- Funnels 39,012 → 600 candidates
- Execution time: 22.2s

Tier 3: Full Scoring

- Complete 6-layer precision scorer & explainer
- Funnels 600 → top 100 shortlist
- Execution time: 1.4s

LAYER 0 — PROTECTION

To protect the ranking shortlist from synthetic, paradoxical, or corrupted profiles, NEXUS executes 6 independent statistical and timeline integrity checks.

Temporal & Paradox Audits

- Identifies start/end date inconsistencies
- Detects 'expert' skills claimed with 0 months experience

Title-Career Mismatch

- Flags profiles claiming senior ML titles but having purely non-technical histories

Skills & Uniformity checks

- Penalizes implausible skill counts (10+ experts)
- Flags suspiciously uniform exam scores

LAYER 2 – CONTEXTUAL FIT

A candidate's suitability is not the sum of their keyword matches. NEXUS inspects the chronological structure of their history to assess quality and growth.

Product Company Focus

Calculates the proportion of career months spent at product organizations versus services/consulting firms (TCS, Wipro, etc.). Product engineers are prioritized.

Keyword Stuffing Filter

Compares claimed skills against actual career description logs. If a candidate claims a skill but has zero mention in their career history, their score is penalized.

Growth & Education

Evaluates rank advancement over time (IC growth) and parses school/major relevance to estimate educational prestige.

LAYER 4 – CANDIDATE INTENT

A perfect candidate is of no use to a recruiter if they are unresponsive. NEXUS groups 23 behavioral indicators into 5 composite indices used as a final score multiplier.

Indices 1-3

- Availability: Login freq & job-seeking status
- Engagement: Latency & message response rate
- Location Fit: Relocation & geo-match

Indices 4-5

- Trust: Verified identity & phone validation
- Capability: Coding score & code quality metrics

Multiplicative Modifier

Behavioral scores act as a multiplier (0.50x to 1.20x) instead of additive, ensuring disengaged candidates naturally drop in rank.

LAYER 5 — EXPLANATIONS

Recruiters manually inspect candidate justifications. NEXUS generates detailed, dynamic, non-templated reasoning explanations matching candidate attributes to JD requirements.

Verifiable Justifications

Explanations refer to actual candidate attributes (years of experience, target skills, current title, and engagement stats) to avoid hallucinations.

Tone & Rank Consistency

The reasoning text is calibrated to match the rank position — glowing for top candidates, critical for low-tier fits.

Example (Rank 1)

'Senior AI Engineer with 7 years experience building RAG systems at product companies; strong recent engagement (100% response rate) and Pune-based.'

SUBMISSION STATUS

NEXUS meets all rules, schemas, and performance criteria of the Data & AI Challenge.

✓ Submission CSV

Exactly 100 rows, headers match schema, ranks sorted, validated via script with zero errors. Output matches the team_xxx.csv guidelines.

✓ Sandbox UI

Stunning interactive Streamlit application (demo/app.py) for visualization, allowing immediate execution on small-sample uploads.

✓ Code Repository

Clean, complete GitHub repo with 5 distinct commits demonstrating Git history authenticity and real development iteration.